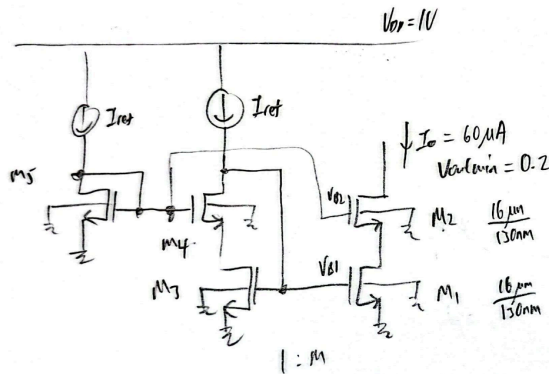


Chai Zi Yang A0262349Y EE5507 Tut 2

1. Hand Calculations



$$I_D = \frac{1}{2} k_n' \left(\frac{W}{L}\right) V_{ov}^2$$

$$V_{ov} = \sqrt{\frac{60\mu A}{\frac{1}{2} \times 100\mu A \times \frac{16\mu m}{130nm}}}$$

$$\approx 0.099V$$

$\Rightarrow V_{out, min}$ of 2V is already satisfied by M_1, M_2 sizing given

$$V_{THN} = 0.4V \quad \mu_n C_{ox} = k_n' = 100\mu A/V^2 \quad \lambda_n = 0.61V^{-1}$$

$$W, L > 130nm \quad 2I_{ref} < 5\mu A$$

Let $I_{ref} = 2.4\mu A$ (total = $4.8\mu A < 5\mu A$)

$$M = \frac{60\mu A}{2.4\mu A} = 25$$

$$\Rightarrow (W/L)_3 = (W/L)_4 = \frac{16\mu m}{130nm} \div 25 \approx \frac{640nm}{130nm} \leftarrow \text{Keep } L \text{ the same so } \lambda_1 \approx \lambda_3 \text{ to reduce error}$$

High swing cascode I $\Rightarrow (W/L)_5 = \frac{1}{4} (W/L)_3$

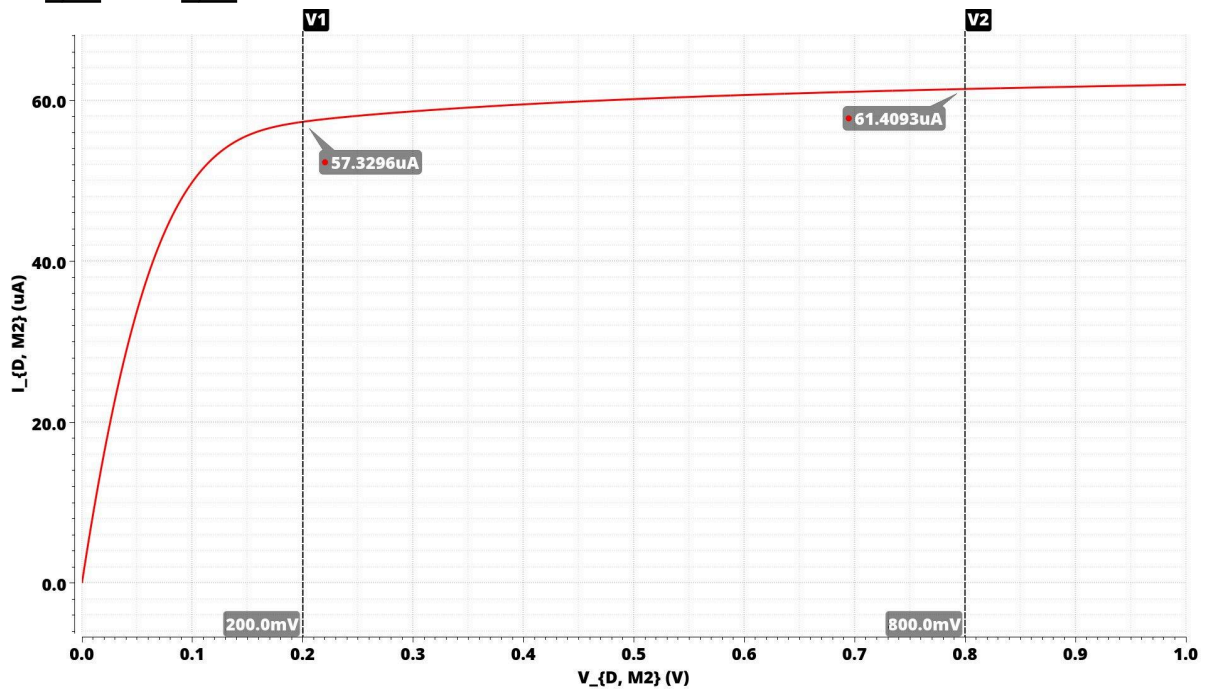
$$= \frac{160nm}{130nm}$$

Let M_5 be the unit transistor

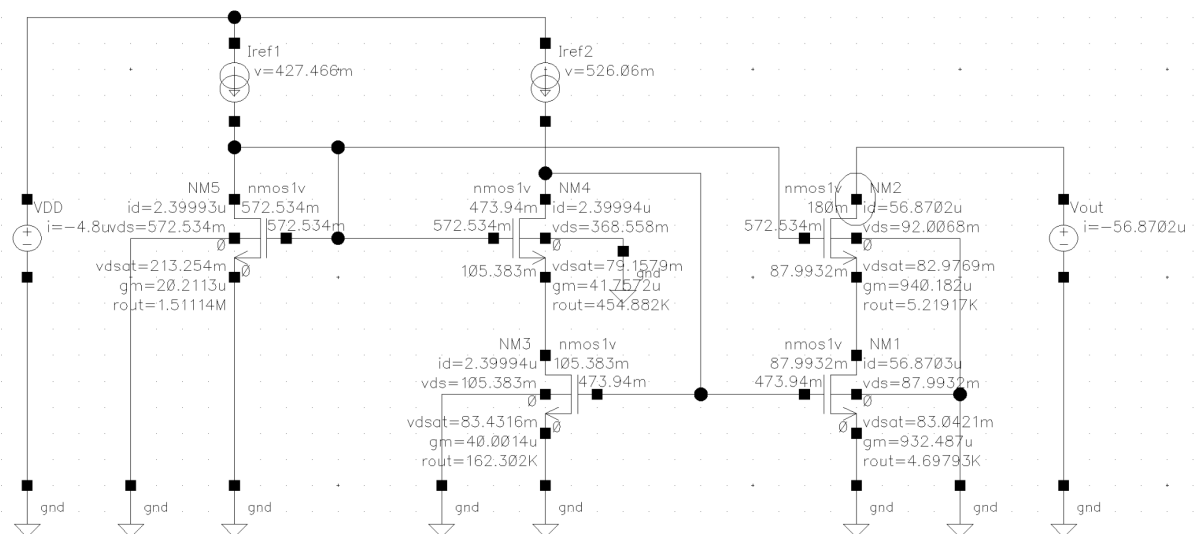
$$\Rightarrow (W/L)_5 = \frac{160nm}{130nm} \quad , \quad (W/L)_3 = (W/L)_4 = 4 \times \frac{160nm}{130nm} \quad , \quad (W/L)_1 = (W/L)_2 = 100 \times \frac{160nm}{130nm}$$

multiplication

2. $I_{D, M2}$ vs $V_{D, M2}$

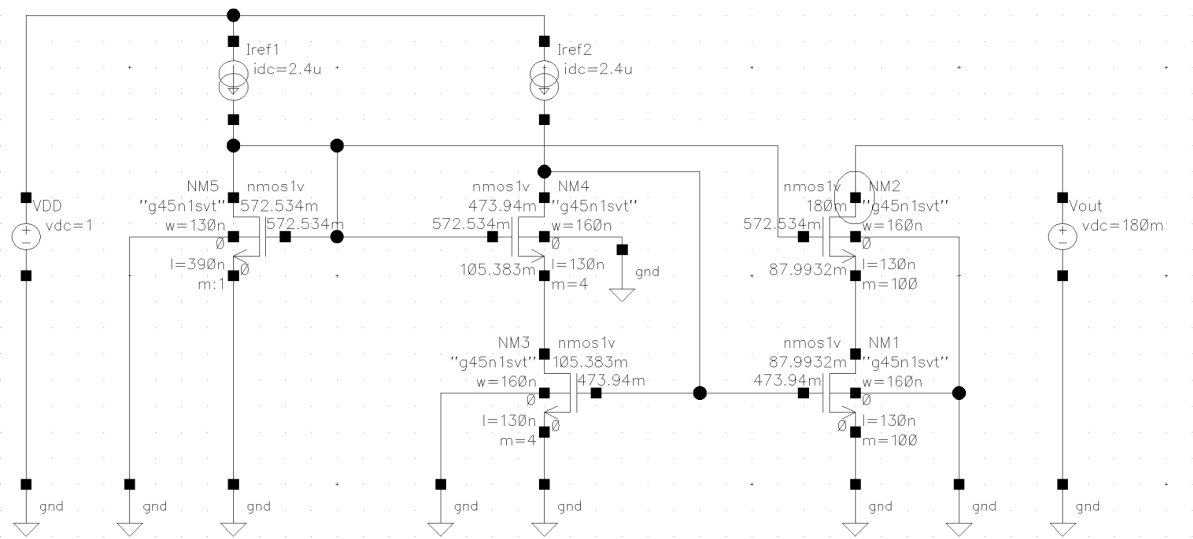


3. Final Schematic Showing DC Operating Point



DC operating point shown here is the worse case Vout with some margin, $V_{out} = 0.18\text{ V}$

4. Final Schematic Showing Sizing



Final schematic defer with hand calculation only by the $(W/L)_{M5}$. $(W/L)_{M5}$ was reduced while I_{ref} was maintained at $2.4 \mu A$ to increase V_{GS5} . $(W/L)_{M5}$ was chosen to be $130nm/390nm$ to ensure NM1 and NM2 are both safely in the saturation region even under the case of $V_{out} = 0.18 V$